Today's Work · Wednesday, October 28

Module 2, Topic 2, Lesson 3: Perpendicular Lines · Exit Ticket on Lessons 1–3 today

Name _____ Period _____

1 · Lesson 3 video (4 minutes): fill in while you watch

When a line turns 90°, its new slope is the _____ _____ of the old slope.

In my own words: **flip** the fraction and _____ .

The slopes of two perpendicular lines multiply to _____ .

Slope of a line	2/3	−4	1	−5/2	7
Perpendicular slope					
Parallel slope					

2 · Horizontal and vertical lines: HOY VUX

Horizontal line: slope is _____ , equation **y =** _____ **V**ertical line: slope is _____ , equation **x =** _____

3 · Book problems (Student Edition): check off as you finish

Problem	**Skill (exit ticket question)**
☐ Activity 3.1 · Worked Example, p. 308 and Question 4, p. 309	Perpendicular slopes (Q7)
☐ Activity 3.2 · Questions 6–10, p. 311	Horizontal and vertical lines (Q3)
☐ Activity 2.1 · Question 6, p. 292	Moving left and right (Q4, Q5)
☐ Activity 2.3 · one graph, p. 298	a·f(x) and f(bx) (Q1, Q6)
☐ Activity 3.3 · Questions 3–4, pp. 312–313	Write a perpendicular line's equation (Q7, Q8)

4 · Before the exit ticket: remember

- **Outside** the parentheses moves up/down: f(x) + 3 is up 3. **Inside** moves left/right, backwards: f(x − 3) is **right** 3, f(x + 3) is **left** 3.
- With f(x) = x: a · f(x) = ax and f(bx) = bx, so both just change the **slope**. Line falls left to right → negative slope.
- Table shifted right 3: the new y-intercept comes from x = −3 (which x lands on 0?).
- **Parallel** → same slope. **Perpendicular** → flip and change the sign. "Crosses the x-axis at 6" means the point **(6, 0)**.
- Point-slope: $y - y_1 = m(x - x_1)$. Watch signs: x − (−2) = x + 2.
- **Range** = the y-values. The **zero** of a function = where it crosses the x-axis.

Homework tonight: Lesson 2 quiz on the class website. Stuck? Rewatch the Lesson 1, 2 or 3 video.